



# International Islamic University Chittagong

## Morality Development Program

Semester End Examination, Semester: Autumn - 2023

Course Title: Concepts of Moral Development-I, Course Code: MDP-2404

[Other than Shariah Faculty]

Full Marks: 50

Time: 2.5 Hours

*[N.B: Answer the following questions]*

1. Describe the meaning of al-ma'roof and al-munkar. Cite the importance of enjoining good and forbidding evil. 10  
  
OR  
  
What exactly is a good health? Draw health guidelines from the Quran and Sunnah. 10
2. What is the definition of gambling? Draw various events of gambling which are established in our state and society. State along with their adverse effects in our life. 10  
  
OR  
  
Define the term "Leadership". Illustrate the Qualities of a good leader in Islam. 10
3. Explain about Peaceful Co-existence from Islamic Perspective. What are the importance of tolerance and peaceful Co-existence? 10
4. Describe the status of the defender of the Islam in the light of hadith number 52 of Sahih al-Bukhari. 10
5. a) "Be kind, honorable and humble to one's parents"- explain with the holy Quranic verses from Sura al Isra. 5 X 2 = 10  
b) "So establish weight with justice and fall not short in the balance"- deduce the statement in the light of Sura Ar Rahman (Ayat Number-09).



**International Islamic University Chittagong (IIUC)**  
**Department of Computer Science and Engineering (CSE)**  
**Semester Final Examination**

|                        |                              |
|------------------------|------------------------------|
| Program: B. Sc. in CSE | Semester: Autumn-2023        |
| Course Code: MATH-2407 | Course Title: Mathematics-IV |
| Time: 2:30 hours       | Total Marks: 50              |

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.  
(ii) Separate answer script must be used for separate group.  
(iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

**Course Learning Outcomes (CLOs) of the Questions**

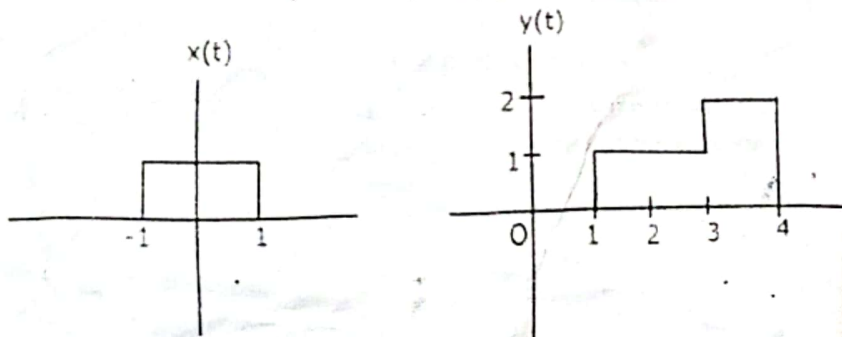
|      |                                                                                                                                                                                                      |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLO1 | Demonstrate the understanding of the basic principles and operations set theory, complex numbers, geometrical interpretation complex functions and the concept of transformation in a complex plane. |
| CLO2 | Apply the concept of transformation of an object into complex space and operation of complex functions                                                                                               |
| CLO3 | Use Fourier series, Laplace's Transforms, Fourier Transform in different scenario.                                                                                                                   |
| CLO4 | Analyze the harmonics & spectrum of different types of waves.                                                                                                                                        |
| CLO5 | Demonstrate the harmonic analysis using MATLAB.                                                                                                                                                      |

**Bloom's Taxonomy Domain Levels of the Questions**

| Letter Symbols | R        | U          | Ap    | An      | E        | C      |
|----------------|----------|------------|-------|---------|----------|--------|
| Meaning        | Remember | Understand | Apply | Analyze | Evaluate | Create |

**Group-A**

- |       |                                                                                                                                                                                                                                                                                                          | Mark | CLO  | DL |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|----|
| 1.    | If $y = f(t) = t^2$ over the interval $-\pi < t < \pi$ and has period $2\pi$ . Here, $T = 2L = 2\pi \therefore L = \pi$                                                                                                                                                                                  |      |      |    |
| a)    | Find the Fourier series for the above function.                                                                                                                                                                                                                                                          | 5    | CLO3 | U  |
| b)    | Hence, deduce $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$                                                                                                                                                                                                 | 3    | CLO3 | U  |
| c)    | Write down the complex form of Fourier series.                                                                                                                                                                                                                                                           | 2    | CLO1 | R  |
| 2. a) | Evaluate the convolution sums of $y[n] = x[n] * h[n]$<br>Where,<br>$x[n] = \begin{cases} 1, & n = 0 \\ 3, & n = 1 \end{cases} \text{ and } h[n] = \begin{cases} 2, & n = 0 \\ 1, & n = 1 \end{cases}$ Where, n represents the time index.<br><b>Or</b><br>Find the convolution integral of $x(t) * y(t)$ | 6    | CLO4 | U  |



- b) i. If  $x = \cos 4t + \frac{1}{3} \sin 4t$  then find the amplitude of the new signal and the phase shift. 2+2=4 CLO4 An

ii. Is there any difference in line spectrum in Fourier series and Fourier transform? What kind of difference? And why?

Or

Plot the line (at least 4) spectrum for the following complex wave 4 CLO4 An

$$f(t) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{1}{2n-1} \cos nt + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} \sin nt$$

### Group-B

3. a) Find the Fourier transform of the function,

$$f(t) = \begin{cases} 1, & 0 < t < 1 \\ -1, & -1 < t < 0 \\ 0, & \text{otherwise} \end{cases}$$

Or

Find the Laplace transform of the functions  $f(t) = \cos at$

Mark  
5 CLO CLO3 DL  
Ap

- b) Express the following function in terms of unit step functions and find its Laplace transform:

$$f(t) = \begin{cases} 10; & t \leq 3 \\ 8; & t \geq 3 \end{cases}$$

Or

Solve the following Initial Value Problem (IVP) by Laplace Transform

$$Y'' + 4Y = 12t \quad Y(0) = 0 \quad Y'(0) = 7$$

5 CLO3 Ap

4. a) Evaluate  $\mathcal{L}\left\{\frac{\sin 2t}{t}\right\}$  using the division theorem.

5 CLO3 Ap

- b) Draw the Impulse Function  $2 * \delta(t - 3)$

1 CLO3 U

- c) Draw the Function  $x(t) = r(t+1) - r(t) + r(t-2)$

4 CLO3 U

5. a) Write down a user defined function in MATLAB to reconstruct  $f(t)$  in the time interval of  $[-4, 20]$  for the following complex wave

5 CLO5 An

$$f(t) = 10 + \frac{10}{3\pi} \left( \sin \frac{\pi t}{4} + \frac{1}{3} \sin \frac{3\pi t}{4} + \frac{1}{5} \sin \frac{5\pi t}{4} + \frac{1}{7} \sin \frac{7\pi t}{4} + \dots \right)$$

- b) Write down MATLAB code for the following function:

3 CLO5 An

$$x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-4)$$

- c) If

2 CLO5 An

$$x[n] = 2; \quad n = 0$$

$$h[n] = 2; \quad n = 0$$

$$= 6; \quad n = 1$$

and

$$= -2; \quad n = 1$$

Write MATLAB code to find the convolution sum of the above signals.

International Islamic University Chittagong  
Department of Computer Science and Engineering  
B. Sc. Engineering in CSE  
Semester Final Exam, Autumn 2023

Course Code: ACC 2401  
Time: 2 hours 30 minutes

Course Title: Financial & Managerial Accounting  
Full Marks: 50

- (i) The figures in the right-hand margin indicate full marks  
(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

| Course Outcomes (COs), Program Outcomes (POs) and Bloom's Levels (BL) of the Questions |                                                                                                                                     |      |
|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------|
| CO                                                                                     | CO Statements                                                                                                                       | PO   |
| CO1                                                                                    | Explain the basic concept of financial accounting, cost accounting and management accounting.                                       | PO11 |
| CO2                                                                                    | Analyze the basic concept of Cost Accounting and preparation of Cost Sheet.                                                         | PO11 |
| CO3                                                                                    | Apply the tools from accounting and cost accounting this would facilitate the decision making i.e. Budgeting, Make or Buy decision. | PO11 |
| CO4                                                                                    | Compare the different business situations and suggest to best solution with analytical abilities.                                   | PO11 |

| Bloom's Levels (BL) of the Questions |          |            |       |         |          |        |
|--------------------------------------|----------|------------|-------|---------|----------|--------|
| Letter Symbols                       | C1       | C2         | C3    | C4      | C5       | C6     |
| Meaning                              | Remember | Understand | Apply | Analyze | Evaluate | Create |

### Part A

[Answer the questions from the followings]

1. An inexperienced book-keeper has drawn up an Incorrect and Unadjusted Trial Balance, You are required to Correct/Adjusted Trial Balance. CO2 C2 10

**Asim Manufacturing Co.**  
Adjusted Trial Balance  
as at 30 June 2023

|                           | Dr (Tk)       | Cr (Tk)       |
|---------------------------|---------------|---------------|
| Bank Account (Cash)       | 19730         |               |
| Receivable                | 5440          |               |
| Equipment                 | 1520          |               |
| Insurance Prepayment      | 4600          |               |
| Properties                | 20000         |               |
| Office equipment          | 3600          |               |
| Accumulated Depreciation  |               | 600           |
| Payable                   |               | 1800          |
| Unearned interest revenue |               | 480           |
| Notes payable             |               | 15000         |
| Capital                   |               | 30000         |
| Drawings                  | 8000          |               |
| Interest revenue          |               | 33680         |
| Salary expense            | 9050          |               |
| Rental expense            | 3200          |               |
| Utility expense           | 1970          |               |
| Supplies expense          | 4080          |               |
| Sundry expense            | 910           |               |
| Interest expense          | 200           |               |
| Rental revenue            |               | 240           |
| Salary payable            |               | 500           |
| <b>Total</b>              | <b>82,300</b> | <b>82,300</b> |

**Adjustment data:**

- (i) Equipment on hand at year end Tk. 200.
- (ii) Depreciation Office equipment, Tk. 200
- (iii) Salaries owed but not yet paid, Tk. 5000
- (iv) Accrued interest revenue , Tk. 1300.
- (v) Tk. 300 of the unearned service revenue has been earned.

**OR**

1.

Welwash (Pvt.) Ltd. sells washing machines for outright cash as well as on hire-purchase basis. The cost of a washing machine to the company is Rs. 10,500. The company has fixed cash price of the machine at Rs. 12,300 and hire-purchase price at Rs. 13,500 payable as to Rs. 1,500 down and the Trial balance at December 31, 2022 as given follows:

CO2 C4 10

**Welwash (Pvt.) Ltd**  
Trial balance  
December 31, 2022

| Particular                          | Dr (Rs.) | Cr (Rs.) |
|-------------------------------------|----------|----------|
| Cash                                | 6000     |          |
| Accounts receivable                 | 5000     |          |
| Supplies                            | 1000     |          |
| Furniture                           | 10000    |          |
| Accumulated depreciation- Furniture |          | 4000     |
| Building                            | 50000    |          |
| Accumulated depreciation- Building  |          | 30000    |
| Account payable                     |          | 2000     |
| Unearned Service revenue            |          | 8000     |
| Salary payable                      |          | 3000     |
| Common Stock                        |          | 10000    |
| Retained Earnings                   |          | 2000     |
| Drawings                            | 25000    |          |
| Service revenue                     |          | 60000    |
| Salary expense                      | 16000    |          |
| Supplies expense                    |          |          |
| Advertising Expense                 | 3000     |          |
| Depreciation expense- Furniture     | 1000     |          |
| Depreciation expense- Building      | 2000     |          |
| Total                               | 119000   | 119000   |

**Required:**

- 1) Prepare Financial Statement for Welwash (Pvt.) Ltd

2.

**TTT Tailoring Service**  
Trial Balance  
April 30, 2020

CO3 C4 10

| Accounts Title                      | Debit           | Credit          |
|-------------------------------------|-----------------|-----------------|
| Cash                                | 2,370           | -               |
| Accounts Receivable                 | 23,540          | -               |
| Supplies                            | 10,570          | -               |
| Prepaid Insurance                   | 7,660           | -               |
| Equipment                           | 63,930          | -               |
| Accumulated Depreciation- Equipment | -               | 21,730          |
| Building                            | 74,330          | -               |
| Accumulated Depreciation-Building   | -               | 15,050          |
| Accounts Payable                    | -               | 19,550          |
| Interest Payable                    | -               | -               |
| Wage Payable                        | -               | -               |
| Unearned service revenue            | -               | 8,840           |
| N/P-Long Term                       | -               | 69,900          |
| M. Lpoez, Capital                   | -               | 46,200          |
| M. Lopez, Withdrawals               | 47,500          | -               |
| Service revenue                     | -               | 92,170          |
| Depreciation expense-Equipment      | -               | -               |
| Depreciation expense-Building       | -               | -               |
| Wages expense                       | 28,970          | -               |
| Insurance expense                   | -               | -               |
| Interest expense                    | 5,890           | -               |
| Utilities expense                   | 5,670           | -               |
| Property Tax expense                | 3,010           | -               |
| Supplies expense                    | -               | -               |
| <b>Total</b>                        | <b>2,73,440</b> | <b>2,73,440</b> |

Additional data at April 30, 2020:

- Supplies used during the year Tk.6,880.
- Prepaid insurance expired during the year, Tk.5,370.
- Accrued Interest expense Tk.2,280.
- Accrued service revenue Tk.2,200.
- Depreciation for the year: equipment Tk.6,700; building 3,210.
- Accrued wage expense Tk.830.
- Unearned service revenue earned during the year Tk.5,180.

**Requirement:** Prepare adjusting entries and an adjusted trial balance.

### Part B

[3]

CO2 C4 10

Prepare a **Cost of Goods Sold Statement** for **Hisham & Sons Company** based on the following information:

|                                           |          |
|-------------------------------------------|----------|
| Sales discount                            | \$ 1800  |
| Purchase discounts                        | 350      |
| Sales                                     | 2,80,000 |
| Purchase returns and allowances           | 2,150    |
| Depreciation:                             |          |
| --Machinery (three fifth factory related) | 25,000   |
| --Building (two third factory related)    | 18,000   |
| Factory insurance                         | 5,140    |
| Freight out                               | 2800     |

|                                        |        |
|----------------------------------------|--------|
| Other factory expenses                 | 1,600  |
| Production supervisor salary           | 5,000  |
| Land                                   | 50,000 |
| Bond interest expense                  | 5,000  |
| Indirect Materials                     | 2,350  |
| Sales salaries                         | 10,480 |
| Building Rent (80% production related) | 25,000 |
| Freight in                             | 1,500  |
| Direct factory labor                   | 85,500 |
| Materials purchases                    | 42,350 |
| Supplies expense                       | 2,000  |
| Factory Utility expense                | 3,000  |
| Office salaries                        | 10,600 |
| Advertising expenses                   | 1,200  |
| Indirect Labor                         | 4,350  |

|                      |                        |                          |
|----------------------|------------------------|--------------------------|
| <b>Inventories:</b>  | <b>January 1, 2018</b> | <b>December 31, 2018</b> |
| Finished goods.....  | \$6,000                | \$4,000                  |
| Work-in-process..... | 5,000                  | 2,000                    |
| Raw Materials.....   | 10,000                 | 8,000                    |

- [4] Super Sales Company is the exclusive distributor for a revolutionary booking. CO3 C5 10  
The product sells for \$60 per units and variable expenses per unit \$36. The company's fixed expenses are \$360,000 per year.

**Required:**

- What are the Super Sales Company CM ratio?
- Using the equation method:
  - What is the break-even point in units and in sales dollars?
  - What sales level in unit and in sales dollars in required to earn an annual profit of \$90,000?
  - Assume that through negotiation with the manufacturer the Super Sales Company is able to reduce its variable expenses by \$3 per unit. What is the company's new break-even point in units and in sales dollars?
- What are the Super Sales Company MOS and OL?

- [5] MMM Corporation manufactures and sells a seasonal product that has CO3 C4 10  
peak sales in the third quarter.  
The following information concerns operations for Year 2—the coming year—and for the first two quarters of Year 3:

- The company's single product sells for \$10 per unit. Budgeted sales in units for the next six quarters are as follows (all sales are on credit):

|               | Year 2              |                     |                     |                     | Year 3              |                    |
|---------------|---------------------|---------------------|---------------------|---------------------|---------------------|--------------------|
|               | 1 <sup>st</sup> QTR | 2 <sup>nd</sup> QTR | 3 <sup>rd</sup> QTR | 4 <sup>th</sup> QTR | 1 <sup>st</sup> QTR | 2 <sup>nd</sup> QT |
| Budgeted unit | 60,000              | 70,000              | 80,000              | 90,000              | 100,000             | 110,00             |

- Sales are collected in the following pattern: 65% in the quarter the sales are made, and the remaining 35% in the following quarter. On January 1, Year 2, the company's balance sheet showed \$75,000 in accounts receivable, all of which will be collected in the first quarter of the year. Bad debts are negligible and can be ignored.
- The company desires an ending finished goods inventory at the end of each quarter equal to 20% of the budgeted unit sales for the next quarter. On December 31, Year 1, the company had 15,000 units on hand.

**Required:**

Prepare the following budgets and schedules for the year, showing both quarterly and total figures:

- i. A sales budget and a schedule of expected cash collections.
- ii. A production budget.

OR

[5]

Peak sales for Midwest Products, Inc. occur in August. The company's sales budget for the third quarter showing these peak sales is given below:

CO3 C4 10

|                | July      | August    | September | Total       |
|----------------|-----------|-----------|-----------|-------------|
| Budgeted sales | \$600,000 | \$900,000 | \$500,000 | \$2,000,000 |

From past experience, The company has learned that 20% of a month's sales are collected in the month of sale, that another 70% is collected in the month of following sale, and that the remaining 10% is collected in the second month following sale. Bad debts are negligible and can be ignored. May sales totaled \$430,000, and June sales totaled \$540,000.

**Required:**

- a. Prepare a schedule of expected cash collections from sales. by month and in total, for the third quarter.
- b. Assume that the company will prepare a budgeted balance sheet as of September 30. Compute the accounts receivable as of that date.

# International Islamic University Chittagong

## Department of Computer Science and Engineering

B. Sc. in CSE Semester Final Examination, Autumn 2023

Course Code: CSE 2421

Course Title: Computer Algorithms

Time: 2 Hours 30 Minutes

Total Marks: 50

### Group A

[Answer the following questions]

1. a) Given arrival and departure times of all trains that reach a railway station, produce an algorithm that finds the minimum number of platforms required for the railway station so that no train waits. Illustrate with the following input:

arrival[] = {9:00, 9:40, 9:50, 11:00, 15:00, 18:00}  
departure[] = {9:10, 12:00, 11:20, 11:30, 19:00, 20:00}

Or

1. a) Suppose we have a knapsack that has a weight limit  $w$ . There are items  $i_1, i_2, \dots, i_n$ , having weights  $w_1, w_2, \dots, w_n$ , respectively, and associated benefits  $v_1, v_2, \dots, v_n$ , in that order. We need to maximize the total benefit such that the total weight inside the knapsack is at most  $w$ . Demonstrate an algorithm to solve this problem.
1. b) Construct the Huffman codes for encoding the text 'the more you read, the more you learn'. Considering the letters present and their associated frequencies, draw the corresponding Huffman tree and utilize the tree to yield the binary encoding for each letter.
2. a) Consider the following graph in Fig.2(a). In what order will the nodes be visited using a Breadth First Search? In what order will the nodes be visited using a Depth First Search? Assume A to be the source. If there is ever a choice to be made between multiple neighboring nodes in the BFS or DFS algorithms, decide the orders alphabetically.

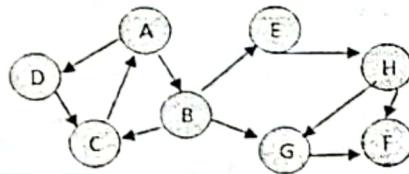
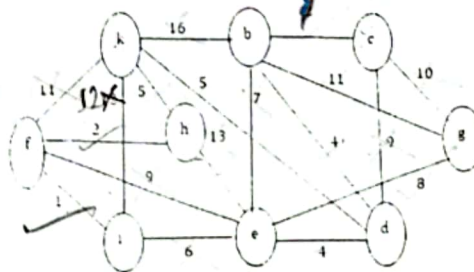


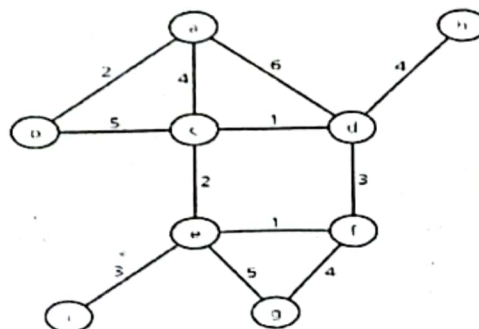
Fig. 2(a)

2. b) Consider the following graph. Showing the steps, construct a minimum spanning tree using Kruskal algorithm.



Or

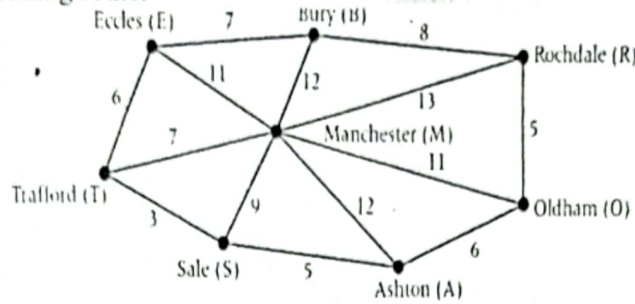
2. b) Find a minimum spanning tree for the following graph using Prim's algorithm and show the steps in figure.



### Group B

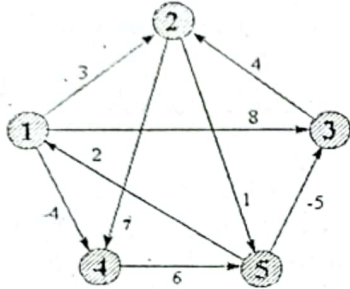
[Answer the following questions]

3. a) The following diagram shows main roads connecting places near Manchester, where the values labelling the edges represent the distances in miles. Mark lives in Rochdale and works in Trafford. Use *Bellman-Ford's* algorithm to find the shortest distance from Rochdale to Trafford. Write down the corresponding route.



Or

3. a) Consider the following graph for finding all pair shortest path using Floyd-Warshall algorithm.



$D(4) =$

|   |    |    |   |    |
|---|----|----|---|----|
| 0 | 3  | -1 | 4 | -4 |
| 3 | 0  | -4 | 1 | -1 |
| 7 | 4  | 0  | 5 | 3  |
| 2 | -1 | -5 | 0 | -2 |
| 8 | 5  | 1  | 6 | 0  |

What is the value of matrix  $D(5)$  calculated from matrix  $D(4)$  given above? What is key the difference between the Floyd-Warshall algorithm and the Dijkstra's algorithm?

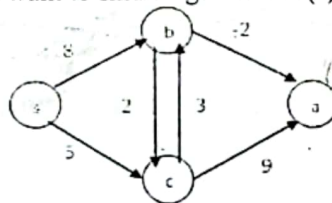
3. b) i. Consider the following cases in a graph

Case 1: The weights of the edge  $\langle e, f \rangle$  and edge  $\langle f, e \rangle$  are  $-2$  and  $7$  respectively.

Case 2: The weights of the edge  $\langle e, f \rangle$  and edge  $\langle f, e \rangle$  are  $2$  and  $-7$  respectively.

Which one does form a negative weight cycle and how?

- ii. Consider following graph. We want to find single source (s) shortest path for this graph.



- I. Which algorithm should you apply?
- II. Let us change the weight of the edge  $\langle b, a \rangle$  to  $+2$ . Now what algorithm should you use? Find the single source shortest path for the case II above. You are recommended to apply the fastest possible greedy algorithm. Show the steps.

4. a) How do you determine whether two lines on a plane intersect or not? Explain with an example.
4. b) Perform Graham's scan on the following set of points to find their *convex hull* (x and y coordinates of each point in brackets):

$p_0(1,1), p_1(3,3), p_2(3,5), p_3(1,2), p_4(4,4), p_5(2,4), p_6(3,1), p_7(5,3).$

Or

4. b) Write down the Euclid's algorithm to determine GCD. What is computational geometry and what are its areas of application?

5. a) Clearly define the following terms.

(i) *Polynomial time algorithm* (ii) *Decision problem* (iii) *Optimization problem*  
(iv) *Intractable problem* (v) *Tractable problem* (vi) *Deterministic machine*

5. b) With appropriate figures, show a backtracking or recursive solution to the 4-queens problem.

5. c) With a diagram, show the relationship among the following complexity classes: P, NP, NP-Complete and NP-Hard. Are any parts of it still formally unresolved?

**International Islamic University Chittagong**  
**Center for General Education (CGED)**  
**Semester End Examination Autumn-2023**  
Course Code: GEGL-2401  
Course Title: Bangla Language and Literature  
Full Marks: 50 Time: 2.5 Hours

ক-বিভাগ  
ভাষা ও নিমিত্তি: ৩০  
(প্রতিটি প্রশ্নের মান সমান।)

| প্রশ্ন নং | বর্ণনা                                                                                                                                                     | মান | CLO  | Cognitive learning |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|--------------------|
| ০১.       | ক. নিচের শব্দগুলো থেকে ভুল-শুদ্ধ বানান পার্থক্য কর:<br>শৃংখলা, প্রতিযোগী, নিঃস্থান, ক্ষিদে, কার্যক্রম, দুজন,<br>অহঙ্কার, মাস্টার্স, স্থাপত্য বিদ্যা, তবুও। | ৫   | CLO1 | Apply              |
|           | খ. পাঁচটি বাংলা যতিচিহ্নের প্রয়োগসহ ব্যবহার আলোচনা কর।                                                                                                    | ৫   | CLO1 | Understand         |
| ০২.       | “ডেপু প্রতিরোধে চাই কার্যকরী পদক্ষেপ”-শীর্ষক পত্রিকায়<br>প্রকাশের উপযোগী একটি পত্র রচনা কর।                                                               | ১০  | CLO1 | Create             |
| ০৩.       | একটি স্বরচিত খুদে গল্প উপস্থাপন কর।                                                                                                                        | ১০  | CLO1 | Create             |
|           | অথবা,<br>সংক্ষিপ্ত আলোচনা কর:<br>ক. আধুনিক তথ্য-প্রযুক্তি। খ. মানবতা ও নৈতিকতা।                                                                            | ১০  | CLO1 | Create             |

খ-বিভাগ  
সাহিত্য: ২০

| প্রশ্ন নং | বর্ণনা                                                                                        | মান | CLO          | Cognitive learning |
|-----------|-----------------------------------------------------------------------------------------------|-----|--------------|--------------------|
| ০১.       | “মাতৃ-ভাষা-রূপে খনি, পূর্ণ মণিজালে”-পংক্তির আলোকে<br>“বঙ্গভাষা” কবিতার প্রতিপাদ্য আলোকপাত কর। | ১০  | CLO2<br>CLO3 | Evaluate           |
|           | অথবা,<br>“যৌবনে দাও রাজটিকা” প্রবন্ধের মূলবক্তব্য বিশ্লেষণ কর।                                | ১০  | CLO2<br>CLO3 | Analyze            |
| ০২.       | ভাষা আন্দোলনের প্রতিবাদী শিল্পভাষ্য “কবর”-উক্তির যথার্থতা<br>নির্ণয় কর।                      | ১০  | CLO3         | Evaluate           |

# International Islamic University Chittagong

## Department of Computer Science and Engineering

B. Sc. in CSE Semester Final Examination, Autumn 2023

Course Code: CSE 2423 Course Title: Database Management System

Time: 2 hours 30 minutes

Total marks: 50

[Answer the following questions. Figures in the right hand margin indicate full marks.]

CLO DL

### Group A

[Answer the following questions]

1. a) Describe encryption decryption and privilege. 2 CLO1 C2  
b) Given the following schema 3 CLO1 C2  
Client(cid, cname)  
Contract(cn\_id, client\_id, contract\_title, dateFrom, dateTo)  
Write an assertion that will allow only one contract for a client from "19-Jun-2020" to "25-Jun-2020".  
c) Given the following table of information 5 CLO1 C2  
Employee (ID, cname, Age, Address, Salary, designation, dept)  
Salary\_pay(pid, ID, salary\_of\_month, pay\_date)  
Write a trigger that will automatically increase the salary (5%) of the employee if he is from "finance" department and receiving the salary of June 2020 and onwards.

### OR

- c) What are the different types of referential integrity constraints, and how can they be used to ensure the integrity of data in a relational database? 5 CLO1 C2  
2. a) Explain the problem (with examples) if the data remain un-normalized. How could you resolve them? Explain with example 5 CLO2 C3

### OR

- a) What is an equivalence of sets of functional dependencies? 5 CLO2 C3

$R = (A, B, C, D, E, F)$

$F1 = \{A \rightarrow BC, B \rightarrow CDE, AE \rightarrow F\}$

$F2 = \{A \rightarrow BCF, B \rightarrow DE, E \rightarrow AB\}$

Check whether F1 and F2 are equivalent or not.

- b) The following relation schema can be used to register information on the repayments on micro loans. 5 CLO2 C2

**Repayment (borrower\_id, name, address, loanamount, requestdate, repayment\_date, repayment\_amount)**

A borrower is identified with an unique borrower\_id, and has only one address. Borrowers can have multiple simultaneous loans, but they always have different request dates. The borrower can make multiple repayments on the same day, but not more than one repayment per loan per day.

- a) State a key (candidate key) for Repayment.  
b) Make the normalization to BCNF. Show the steps.

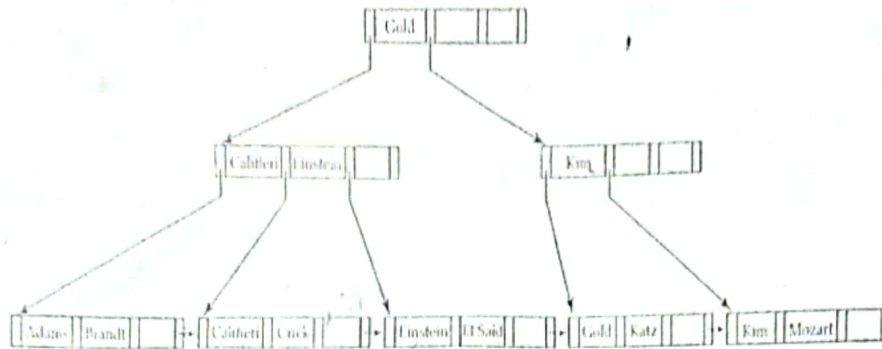
### Group B

[Answer the following questions]

3. a) What is the difference between closed hashing and open hashing in DBMS, and which one is generally more efficient? 2 CLO1 C2

- b) Given the following B+-tree, do the following in it (Step by step):  
 Insert: Marry, Hasi, Nicolous, Oli, Panalous, Quantas, Russel, Stain, Trucer, Zena  
 Delete: Gold, Einstein, Kim, Russel, Stain, Trucer, Zena

5 CLO1 C2



- c) What are the advantages and disadvantages of using hash structures?

3 CLO1 C2

OR

- c) Construct a B+ tree for the following set of data: [5, 10, 15, 20, 25, 30, 35, 40, 45, 50]

4. Answer the following questions:

10 CLO1 C2

1. List the ACID properties. Explain the usefulness of each
2. Justify the following statement: Concurrent execution of transactions is more important when data must be fetched from (slow) disk or when transactions are long, and is less important when data is in memory and transactions are very short.
3. Explain the distinction between the terms *serial schedule* and *serializable schedule*
4. Since every conflict-serializable schedule is view serializable, why do we emphasize conflict serializability rather than view serializability?

OR

Answer the following questions:

10 CLO1 C2

1. Why is cascadelessness of schedules desirable? Are there any circumstances under which it would be desirable to allow noncascadeless schedules? Explain your answer
  2. If deadlock is avoided by deadlock avoidance schemes, is starvation still possible? Explain your answer.
  3. For each of the following protocols, describe aspects of practical applications that would lead you to suggest using the protocol, and aspects that would suggest not using the protocol:
    - Two-phase locking
    - The tree protocol
  4. When a transaction is rolled back under timestamp ordering, it is assigned a new timestamp. Why can it not simply keep its old timestamp?
5. a) Explain the purpose of the checkpoint mechanism. How often should checkpoints be performed? How does the frequency of checkpoints affect
- System performance when no failure occurs
  - The time it takes to recover from a system crash
- b) When is it useful to have replication or fragmentation of data? Explain your answer.
- c) What are the advantages and disadvantages of using multilevel indexing in databases?
- d) What are the advantages of using a B+ tree index over a binary search tree index in a database?

4 CLO1 C2

2 CLO1 C2

2 CLO1 C2

2 CLO3 C2

Or

- d) What are the advantages of using a B+ tree index over a B-tree index for a database?

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG  
 Dept. of Computer Science & Engineering  
 B.Sc. in CSE Semester Final Examination, Autumn 2023  
 Course Code: CSE-2425 Course Title: **Theory of Computing**  
 Full Marks: 50 Time: 2 Hours 30 minutes

[Answer the questions from Group-A and Group-B, using separate answer script.]

**Group – A [20 Marks]**

1. a) Define context-free grammar (CFG). Consider the following CFG:

$$\begin{aligned} G_1: S &\rightarrow aABe \\ A &\rightarrow Abc|b \\ B &\rightarrow d \end{aligned}$$

- i) Identify the elements/components of  $G_1$ , when formally denoted as a 4-tuple.
- ii) Give the leftmost derivation for the string: "abbcbcd".
- iii) Construct the parse tree with  $G_1$  for the input string: "abbcbcd"

Or

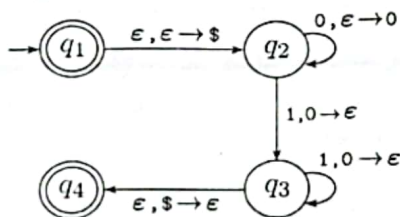
What do you mean by ambiguity? Consider the following CFG:

$$\begin{aligned} G_2: S &\rightarrow E \\ E &\rightarrow E + E | E - E | E * E \\ E &\rightarrow (E) | a \end{aligned}$$

- i) Is this grammar  $G_2$  ambiguous? If yes, prove that using an appropriate string belonging to the language  $L(G_2)$ .
  - ii) Describe the language  $L(G_2)$  in English.
  - iii) Give the rightmost derivation for the input string: " $a + a * (a - a)$ "
- b) Find the CFGs that generate the following languages over the alphabet  $\Sigma = \{0,1\}$
- i)  $L_1 = \{w \mid \text{length of } w \text{ is odd and its middle symbol is '1'}\}$
  - ii)  $L_2 = \{w \mid w = w^R, \text{ that is, the string } w \text{ is a palindrome.}\}$
- c) What is Chomsky normal form (CNF)? Transform the following CFG into an equivalent CFG in CNF:

$$\begin{aligned} G_3: S &\rightarrow ASA | aB \\ A &\rightarrow B | S \\ B &\rightarrow b | \epsilon \end{aligned}$$

2. a) Explain the general notion of pushdown automata (PDA) with the schematic diagram and present its formal definition as a tuple.
- b) Consider the following state diagram for the PDA  $P_1$ :



- i) Identify the elements of this PDA  $P_1$ , denoted as a 6-tuple.
- ii) Give a concise description in English of the language  $L$  that is recognized by the above PDA  $P_1$ . Is this language  $L_3$  regular?
- iii) Is the PDA  $P_1$  deterministic or nondeterministic? Why?

Or

Construct a PDA that recognizes the following language

$$L = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i + j = k\}$$

Give an informal description of the PDA, and illustrate how it proceeds with the input string "abbccc".

- c) State the **pumping lemma** for context-free languages. Using this version of the **pumping lemma** show which of the following language(s) is (are) not context free.

$$\begin{aligned} L &= \{a^n b^m c^n d^m \mid n, m \geq 0\} \\ L &= \{ww \mid w \in \{0, 1\}^*\} \end{aligned}$$

d) For the following CFG, show the state diagram of an equivalent PDA.

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow a$

Or

Remove  $\epsilon$ -production from the following CFG:

$S \rightarrow aB \mid bA$

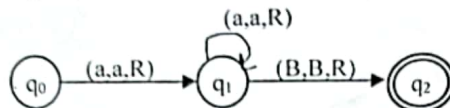
$A \rightarrow Bb \mid a$

$B \rightarrow b \mid \epsilon$

### Group – B [30 Marks]

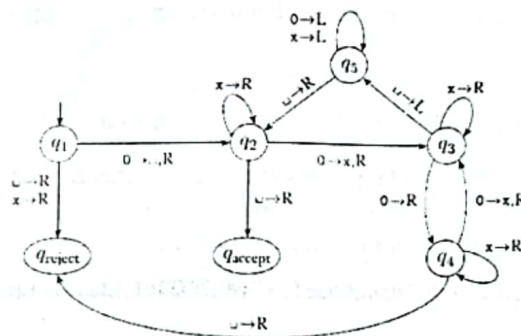
3. a) Explain the notion of the Turing machine (TM) with the schematic diagram and introduce the elements of the tuple that formally defines it. Differentiate between PDA and TM.

b) Give the language  $L$  for the TM represented by the following state diagram:



Is  $L$  a regular language?

c) Consider the following Turing Machine.



Give the sequence of configurations that the machine enters when started with the following strings. i. 00 ii. 000000

Or

Give the implementation-level description of Turing Machines that decide the following languages (any two)

i.  $L = \{w \mid w \text{ is a string with 0s and 1s and contains 1s in a multiple of 3}\}$

ii.  $L = \{ww^R \mid w \text{ is any string of 0's and 1's}\}$

iii.  $L = \{a^n b^n a^n \mid n \geq 0\}$

4. a) Let the language  $A_{DFA}$  be described as  $A_{DFA} = \{\langle B, w \rangle \mid B \text{ is a DFA that accepts input string } w\}$ . Prove that  $A_{DFA}$  is a decidable language.

Or

Let the acceptance language of NFA  $A_{NFA} = \{\langle N, w \rangle \mid N \text{ is an NFA that accepts input string } w\}$ . Show that  $A_{NFA}$  is decidable.

b) Introduce the diagonalization method/argument. Produce an illustrative example/application of it to show that the set of real numbers is uncountable.

c) What is decidability? Let a language  $L$  is defined as  $L = \{\langle M, w \rangle \mid M \text{ is a TM and } M \text{ halts on input } w\}$ . Show that  $L$  is undecidable.

5. a) What are the key ideas to formally prove that every non-deterministic TM has an equivalent deterministic TM?

b) What is polynomial-time reducibility?

d) Formally define the classes P, NP, NP-Hard and NP-complete. How can you show that a Problem is NP-complete?

[END]